

PAPER_687: M87* Black Hole Mass Evolution Over Cosmic Time in the UQFF Framework

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: M87MassEvolutionSimulation | CP4 #271

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Domain: SMBH Evolution / Cosmology

Abstract

We simulate the coupled mass evolution of M87* over 13.8 Gyr combining UQFF-modified Bondi accretion $\dot{M}_{\text{Bondi,UQFF}} = \dot{M}_{\text{Bondi}}(\rho_{\text{eff}}/\rho_{\infty})(1 + f_{\text{TRZ}})$, suppressed Hawking evaporation, and Blandford-Znajek jet power in UQFF. Using RK4 integration with $\Delta t = 10^{13}$ s steps, we find that M87* grows by $\sim 15\%$ over a Hubble time, consistent with observations suggesting limited current growth.

Primary UQFF Equation

$$\frac{dM}{dt} = \dot{M}_{\text{Bondi,UQFF}} + \dot{M}_{\text{evap,UQFF}} - \frac{P_{\text{jet,UQFF}}}{c^2}$$

Parameters

Parameter	Value	Description
M_0	$6.5 \times 10^9 M_{\odot}$	Initial M87* mass
ρ_{ISM}	$1.67 \times 10^{-25} \text{ kg/m}^3$	Intracluster medium
T_{ISM}	10^7 K	ICM temperature
T_{Hubble}	13.8 Gyr	Simulation duration

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density

Constant	Value	Description
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "M87MassEvolutionSimulation.h"
```

```
M87MassEvolutionSimulation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import M87MassEvolutionSimulationCalculator

calc = M87MassEvolutionSimulationCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Walsh et al. 2013 (M87 gas dynamics); Russell et al. 2015 (M87 jet); EHT 2019; UQFF Session 173

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